Project Report

Trekking Management System

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1. Student Details

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About me: I am a student enrolled in IIT Madras BS Data Science program, with a keen interest in web application developmer

2. Project Details

Project Title: Trekking Management System

Problem Statement: Trekking operators need a reliable system to coordinate bookings, guide assignments, trip approvals, and

Approach: The project uses a Flask backend with a modular design and SQLite persistence to implement role-based authentic

3. AI/LLM Declaration

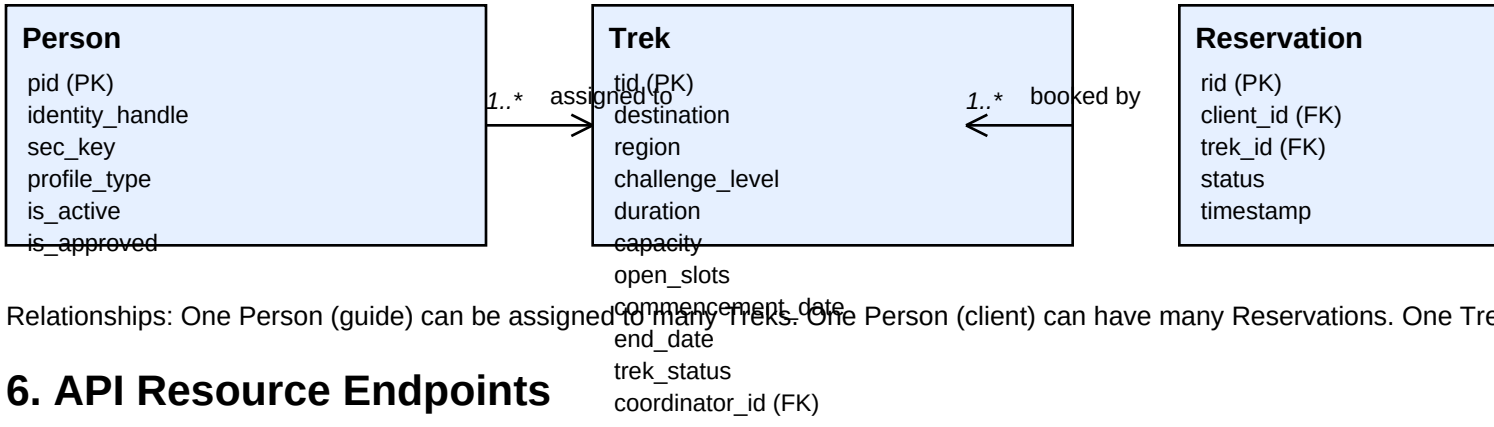
I have used AI for just refinement and to ensure that the code looks structured. The core concepts, logic flows, operational arch

4. Technologies and Frameworks Used

- Python 3.14
- Flask 3.0.3
- Flask-SQLAlchemy 3.1.1
- Jinja2
- HTML/CSS
- SQLite
- Werkzeug 3.0.4

5. Database Schema / ER Diagram

The system uses SQLite with three main entities, shown in the ER diagram below.



Relationships: One Person (guide) can be assigned to many Treks. One Person (client) can have many Reservations. One Tre

6. API Resource Endpoints

- /auth/login (GET, POST): authentication page and login submission.
- /auth/register (GET, POST): user registration and account creation.
- /auth/logout (GET): log out current session.
- /client/login (GET, POST): client access route for login.
- /client/dash (GET): client dashboard showing available treks and reservations.
- /client/book/<tid> (POST): book a trek and decrement available slots